

Econometrics I Mock Exam

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Exam duration: 60 minutes. Total points: 60 (one per minute). Consider the points assigned to each question to manage your time. Read carefully and try to answer precisely. You may use formulas or graphs if you wish. Good luck! **Note: the actual exam will have 90 minutes and 90 points, but roughly the same length as this one (think of having 1/2 more time per question).**

A. Mathematical Questions

(A1) $\mathbb{E}[uu'|X] = \sigma^2 I$ is assumed for obtaining the variances of the OLS estimator.

- i. (3 points) What is the meaning of $\mathbb{E}[uu'|X] = \sigma^2 I$?

Solution: Mathematically:

$$\begin{aligned} V[u|X] &= \mathbb{E}[(u - \mathbb{E}[u])(u - \mathbb{E}[u])'|X] = \mathbb{E}[uu'|X] \\ &= \begin{bmatrix} E[u_1^2|X] & E[u_1 u_2|X] & \cdots & E[u_1 u_n|X] \\ E[u_2 u_1|X] & E[u_2^2|X] & \cdots & E[u_2 u_n|X] \\ \vdots & \vdots & \ddots & \vdots \\ E[u_n u_1|X] & E[u_n u_2|X] & \cdots & E[u_n^2|X] \end{bmatrix} \\ &= \begin{bmatrix} \sigma^2 & 0 & \cdots & 0 \\ 0 & \sigma^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma^2 \end{bmatrix} = \sigma^2 I_{n \times n} \end{aligned}$$

In words: It means that all errors u_i are subject to the same common conditional variance ($\text{var}[u_i|X] = \mathbb{E}[u_i^2|X] = \sigma^2$, $\forall i = 1, \dots, n$) and have no conditional autocorrelation ($\text{cov}[u_i, u_j|X] = \mathbb{E}[u_i u_j|X] = 0$, $\forall j \neq i$).

- ii. (8 points) Provided that $\hat{\beta} = \beta + (X'X)^{-1}X'u$ is unbiased under $y = X\beta + u$ and $\mathbb{E}(u|X) = 0$, how do you arrive at $\mathbb{E}[(\hat{\beta} - \mathbb{E}[\hat{\beta}])(\hat{\beta} - \mathbb{E}[\hat{\beta}])'|X] = \sigma^2(X'X)^{-1}$? Hint: start from $\hat{\beta} = \beta + (X'X)^{-1}X'u$, assume unbiasedness, and plug $\mathbb{E}[uu'|X] = \sigma^2 I$ when needed.

Solution: From the statement we know:

$$\hat{\beta} - \beta = (X'X)^{-1}X'u$$

Now to the variance:

$$\begin{aligned}
V(\hat{\beta}|X) &= \mathbb{E}[(\hat{\beta} - \mathbb{E}[\hat{\beta}])(\hat{\beta} - \mathbb{E}[\hat{\beta}])'|X] \\
&= \mathbb{E}[(\hat{\beta} - \beta)(\hat{\beta} - \beta)'|X] && \text{by unbiasedness} \\
&= \mathbb{E}[(X'X)^{-1}X'u((X'X)^{-1}X'u)'|X] && \text{by statement} \\
&= \mathbb{E}[(X'X)^{-1}X'u(u'X(X'X)^{-1})|X] \\
&\quad \text{(since } (AB)' = B'A') \\
&= (X'X)^{-1}X'\mathbb{E}(uu'|X)X(X'X)^{-1} \\
&= (X'X)^{-1}X'(\sigma^2 I_n)X(X'X)^{-1} && \text{by } \mathbb{E}(uu'|X) = \sigma^2 I_n \\
&= \sigma^2(X'X)^{-1}X'X(X'X)^{-1} \\
&= \sigma^2(X'X)^{-1}
\end{aligned}$$

(A2) You estimate the model $y = X\hat{\beta} + \hat{u}$ through the estimator $\hat{\beta} = (X'X)^{-1}X'y$. However, the data generating process is $y = X\beta + A\gamma + \varepsilon$, with $\mathbb{E}(\varepsilon|X) = 0$.

i. (1 point) What is the technical name of this problem?

Solution: This is a potential omitted variable problem. The regression's error term is $u = A\gamma + \varepsilon$, contains a set of omitted variables $A\gamma$. It potentially violates assumption $\mathbb{E}(u|X) = 0$.

ii. (6 points) Show that your estimate will be unbiased when $\mathbb{E}(X'A|X) = 0$.

Solution:

$$\begin{aligned}
\hat{\beta} &= (X'X)^{-1}X'y \\
&= (X'X)^{-1}X'(X\beta + A\gamma + \varepsilon) \\
&= \beta + (X'X)^{-1}X'A\gamma + (X'X)^{-1}X'\varepsilon \\
\mathbb{E}(\hat{\beta}|X) &= \mathbb{E}[\beta|X] + \mathbb{E}[(X'X)^{-1}X'A\gamma|X] + \mathbb{E}[(X'X)^{-1}X'\varepsilon|X] \\
&= \beta + (X'X)^{-1}\mathbb{E}[X'A|X]\gamma + (X'X)^{-1}\mathbb{E}[X'\varepsilon|X] \\
&= \beta + (X'X)^{-1}\mathbb{E}[X'A|X]\gamma
\end{aligned}$$

If $\mathbb{E}[X'A|X] = 0$, we get bias = $\mathbb{E}(\hat{\beta}|X) - \beta = 0$.

B. **True, ambiguous, or false?** Argue *briefly* whether each of the following statements is true, false, or uncertain based on technical arguments, graphs, and/or formulas.

(B1) (3 points) In the classical regression model, the inclusion of irrelevant variables does not produce any bias but it reduces efficiency.

Solution: True. The absence of a bias can be understood from looking at A2: if $\gamma = 0$, then the bias disappears regardless of whether $\mathbb{E}(X'A|X) = 0$ or $\mathbb{E}(X'A|X) \neq 0$. The variances, however, increase through the $\text{VIF} = \frac{1}{1-R_j^2}$ when the irrelevant variable correlates with other included ones.

- (B2) (3 points) You are estimating an univariate OLS regression and found an observation with a residual that is 10 times higher than all others. Dropping it from the regression will certainly change the slope of your estimate.

Solution: Ambiguous. If that observation happens to have an x_i value close to the mean $\bar{x}$, then the slope will remain rather unaffected. But if x_i is very far away from $\bar{x}$ (a high leverage h_i), then the effect on the slope can be substantial.

- (B3) (3 points) Normally distributed errors are a precondition for the t-statistics to make sense, either with small or infinitely large sample sizes.

Solution: False. It is true that, with small samples, the normality assumption is needed. But, as long as the errors are independently distributed, when $n \rightarrow \infty$, $\hat{\beta}$ will be normally distributed by the central limit theorem. In such case t-statistics will be standard normal and correct.

- (B4) (3 points) In a two-stage least squared instrumental variable estimation, we first regress the treatment of interest against all other remaining regressors, then regress the dependent variable on the resulting residuals.

Solution: False. You regress the treatment against the *instrument* and other remaining regressors, then regress the dependent variable on the resulting *fitted values*.

- C. **Empirical Case Study.** Table 1 shows a series of estimates of the effect of maternal tobacco consumption on newborn birth weight. Answer the following questions based on those results.

- (C1) (4 points) What are $\hat{\beta}_{k \times 1}$ and $\hat{\sigma}^2(X'X)^{-1}_{k \times k}$ according to column (2).

Solution:

$$\hat{\beta} = \begin{bmatrix} 3375.6 \\ -11.4 \\ 89.2 \end{bmatrix} \quad \hat{\sigma}^2(X'X)^{-1} = \begin{bmatrix} 19.96^2 & \text{cov}(\hat{\beta}_1, \hat{\beta}_2) & \text{cov}(\hat{\beta}_1, \hat{\beta}_3) \\ . & 3.236^2 & \text{cov}(\hat{\beta}_2, \hat{\beta}_3) \\ . & . & 27.33^2 \end{bmatrix}$$

- (C2) (5 points) What is the expected weight of a female newborn if the mother is non-smoker? What is the expected 95% confidence interval? Do not compute the value, just state it arithmetically.

Solution: Regression (2) states: Birth weight = 3375.6 – 11.40×cigarettes + 89.2×“Male baby”.

Since cigarettes = 0 and “male baby” = 0, we get: $\widehat{\text{Birth weight}} = 3375.6$ in a 95% confidence interval of $3375.6 \pm 1.96 \cdot 19.96$. Note that 1.96 is a good approximation because $n = 1722$ is high.

- (C3) It is suspected that a lower number of cigarettes is associated with a higher dispersion of birth weights.

- i. (1 point) What result in column (1) may be compromised?

Solution: It invalidates the calculation of the variance-covariance matrix of the estimated coefficients. This invalidates the standard errors and the reported statistical significance.

- ii. (4 points) What two possible remedies would you suggest?

Solution: 1. OLS with Heteroskedasticity-robust standard errors (e.g. White standard errors, with the diagonal of $\widehat{\mathbb{E}(uu|X)}$ corresponding to the residuals squared). 2. A Feasible Generalized Least Square estimation (e.g. a weighted least squared estimation, in which the diagonal of $\widehat{\mathbb{E}(uu|X)} = \hat{\Omega}$ will have the fit of squared residuals on x).

- iii. (4 points) How could you verify this concern using a statistical test?

Solution: We can square the residuals and estimate their linear relationship with the regressor. If the t -value is significantly different from zero, we would be facing heteroskedasticity. Alternatively, we could regress the

$\hat{u}^2$ against a polynomial and perform an F-test, and interpret the results analogously.

- (C4) (4 points) The standard error of the cigarettes coefficient is lower in (2) than in (1), but higher in (3) than in (2). How do you explain this phenomenon?

Solution: The fact that the standard errors are lower in (2) than in (1) reflects the inclusion of a very important variable: the male dummy. The inclusion reduces the variance of the error term σ^2 and thereby leads to a lower estimated variance of the estimated coefficients. This effect predominates over another that is also present: the reduction in efficiency resulting from multicollinearity $\frac{1}{1-R_j^2}$ and the estimation of the error term $\frac{\hat{u}'\hat{u}}{n-k}$ in

$$\widehat{\text{var}}[\hat{\beta}_j|X] = \frac{\hat{u}'\hat{u}}{n-k} \frac{1}{n} \frac{1}{1-R_j^2} \frac{1}{\text{var}[x_j]}$$

In (3) standard errors are larger because of a higher R_j^2 (the variance-inflation bias just explained) and because the number of observations is lower.

- (C5) (4 points) The data of table 1 was produced in a nonexperimental setting. Why should this be a concern for estimating the effect of smoking on birth weights?

Solution: Because the treatment (smoking), being non-random, is possibly correlated with unobservables, such as drinking. Then we may be then misattributing part of the drinking effect as being caused by smoking, for example.

Mathematically, Here smoking is part of X but drinking is part of u . The fact that the treatment x_j correlates with the error term would be a violation of $\mathbb{E}(u|X) = 0$, leading to biased and inconsistent estimates.

- (C6) (4 points) Which of the three estimates seems more appropriate to you for quantifying the impact of a mother consuming one additional cigarette per day on a baby's birth weight? Justify.

Solution: Among the three estimates, that in (3) is the safest bet. Why? Firstly, because the added controls are arguably causal and many of them are likely correlated with smoking (such as age), ameliorating the omitted variable bias explained in answer C5. While it would be desirable to have many more controls (like “drinking”), note that the control “no. of prenatal visits” may at least partially ameliorate the bias of omitted behaviors like “drinking”—caring

mothers are likely having more prenatal visits *and* avoiding drinks.

A second reason suggesting that (3) is better is that it has the largest adjusted R^2 . A large adjusted R^2 does not guarantee a good regression (we could have, for instance, a very large adjusted R^2 in a regression subject to simultaneity, like in the case of crime and police); but, since in (3) all regressors make logical sense, the adjusted R^2 offers further support for choosing that model instead of (1) or (2).

Table 1: OLS regressions

Dependent variable: Birth weight (grams)			
	(1)	(2)	(3)
Constant	3421.7*** (14.15)	3375.6*** (19.96)	2243.4*** (397.2)
Cigarettes per day	-11.48*** (3.245)	-11.40*** (3.236)	-10.30** (3.327)
Male baby (dummy)		89.20** (27.33)	85.09** (27.95)
Mother's age, years			66.51* (27.16)
(Mother's age) ²			-1.050* (0.454)
No. of prenatal visits			11.50** (3.753)
Mother's educ, years			-1.784 (7.167)
Observations	1722	1722	1644
R^2	0.007	0.013	0.023
Adjusted R^2	0.007	0.012	0.019

Standard errors in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$